VIT Vidyalankar Institute of Technology
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Assignment 8

Class	SE CMPN	**Vertical_Basket**	PC_PCC
Course Code	PCCE05T	**Course**	Data Base Management Systems
Announcement Date	23rd September	**Submission Date**	30th September 2025

Self-Declaration

It took me ___1___ hour(s) and ___20___minute(s) to solve the assignment, ensuring accuracy, completeness, and adherence to professional and ethical standards.

Roll No.	24102B0030	**Name**	Shubham Uday Gotad	**Sign**	

Assessment

		Faculty Signature	

Q1. Given a relation schema R(A, B, C, D, E) with functional dependencies:

A.>

1. A → B
2. B → C
3. CD → E

a. Find the attribute closure of A, CD, and AB.

b. Determine all the candidate keys.

B.> Suppose you're given that the closure of attribute set {X} is {A, B, C, D, E}. Propose at least two different sets of functional dependencies that might lead to this closure.

Q2. Given:

R(P, Q, R, S) and F = {P → Q, PQ → R, R → S}

a. Compute P+

b. *Construct a minimal cover* for F

Q3. A relation R(StudentID, CourseID, Instructor, Grade, Department) has the following dependencies:

StudentID, CourseID → Grade

CourseID → Instructor

Instructor → Department

1. a. Identify the anomalies in this schema.

b. Normalize the schema into **3NF**.

c. Can you further normalize it into **BCNF**? Justify your answer.

Q4. Given the following scenario:

A university stores data in a relation R(StudentID, Name, CourseID, CourseName, InstructorName).

a. Assume appropriate dependencies and normalize the relation to **BCNF**.

b. Suggest a real-world situation, where staying in **3NF** might be preferable to BCNF and explain why.

Q5. Create a relation schema with at least four attributes and a set of functional dependencies such that it:

- Is in 1NF
- Violates 2NF and 3NF

Then: Normalize it step-by-step to BCNF with explanation at each step.

*Q6. Design a relation R with a set of FDs such that decomposition into BCNF is **lossless but not dependency-preserving**. Explain the trade-off between normalization and dependency preservation with reference to your example.*

Q7. A hospital maintains a table:

Patient(PatientID, Name, DoctorID, DoctorName, Department, RoomNumber)

- A doctor works in only one department
- A patient is assigned one doctor, but multiple patients may be assigned the same doctor
- Patients can share room numbers

Tasks:

a. Identify the functional dependencies

b. Normalize the table to BCNF

c. Discuss what happens to **data redundancy** and **query complexity** after normalization